



UNIVERSITY OF PERADENIYA - FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) - 2021/22
CSC1023/CS102-Object-oriented Programming/Programming
Techniques (3 Credits)



Answer All Questions

Time allowed: Two Hours

1. [25 Marks]

- a) Explain the scope of the four types of access modifiers available in Java using suitable examples. [6 marks]
- b) Can we use a different name for the “main method” rather than using “main”? Justify your answer. [4 marks]
- c) Explain how robustness can be achieved in Java. [4 marks]
- d) Explain the concept of “abstraction” using a real-world Java example. [6 marks]
- e) What is Java Object class? Discuss one of the important methods of the Object class. [5 marks]

2. [25 Marks]

- a) Discuss the two types of typecasting with regard to objects using an example. [5 marks]
- b) Explain how multiple inheritance is achieved in Java. [4 marks]
- c) “In Java it is possible to have multiple constructors”. Comment on this statement. [4 marks]
- d) Compare and contrast abstract vs. final with respect to methods and classes. [4 marks]
- e) How does garbage collection make Java more memory efficient? [4 marks]
- f) Can we mark a constructor with the static keyword? Explain your answer. [4 marks]

3. [25 Marks]

- a) Explain the polymorphism object-oriented concept using a suitable example from Java collections (i.e., java.util.Collections). [4 marks]
- b) Determine the output of the program segment below. Explain your answer. [4 marks]

```
String st = "Give";  
StringBuilder sb = new StringBuilder("Don't ");  
st.concat("Up");  
sb.append(st);  
System.out.println(sb);
```

- c) Explain the difference between HashSet and HashMap collections. [3 marks]
- d) Show how you would ensure no duplicate elements are added to the Set (Vehicle objects) created below. [4 marks]

```

class Vehicle {
    String number;
    String owner;

    public Vehicle(String number, String owner) {
        this.number = number;
        this.owner = owner;
    }
    public String getNumber() {
        return number;
    }
    public String getOwner() {
        return owner;
    }
}

Set<Vehicle> vehSet = new HashSet<>();
Vehicle s1 = new Vehicle("1001", "Ryan");
Vehicle s2 = new Vehicle("1001", "John");
vehSet.add(s1);
vehSet.add(s2);

```

- e) What is the output of the stuCount variable in the two programs below in i) and ii)?
[4 marks]

i)

```

class Student {
    private static int stuCount = 12;
    public Student() {
        stuCount++;
    }
    public void go() {
        System.out.println(stuCount);
    }
}

public class Main {
    public static void main(
        String[] args) {
        Student s1 = new Student();
        Student s2 = new Student();
        s2.go();
    }
}

```

ii)

```

class Student {
    private int stuCount = 12;
    public void go() {
        stuCount++;
        System.out.println(stuCount);
    }
}

public class Main {
    public static void main(
        String[] args) {
        Student s1 = new Student();
        Student s2 = new Student();
        s1.go();
        s2.go();
    }
}

```

- f) Consider the following class hierarchy and the generic class.

```

class Shape { /* ... */ }
class Circle extends Shape { /* ... */ }
class Rectangle extends Shape { /* ... */ }
class Node<T> { /* ... */ }

```

- i) Will the following code compile? Explain your answer. [2 marks]

```
Node<Circle> nc = new Node<>(Circle);
```
- ii) Write a code segment to create an ArrayList consisting of Circle objects and traverse through it with an Iterator object. [4 marks]

4. [25 Marks]

- a) Explain the concept of chained I/O streams in Java. [4 marks]
- b) The code below utilizes Java I/O streams to read a text file containing bank account information.

```
// Text file: accounts.txt
10001, J Smith, 20000.34
10002, S Campbell, 45300.00
```

```
// Code
class Account {
    private String accno;
    private String accname;
    private double balance;
    // set attributes
    // print account details
}

class ManageAccounts {
    public void readAccounts(String fileName) {
        InputStreamReader isr = new InputStreamReader(new
            FileInputStream(fileName));
        BufferedReader br = new BufferedReader(isr);
        String accno, balance;
        while ((line = br.readLine()) != null) {
            % read the account details to String variables: accno and
              balance
        }
    }
    public void printAccountList() {
        % Print the account list
    }
}

public class AccountApplication {
    public static void main(String[] args) {
        ManageAccounts rff = new ManageAccounts();
        rff.readAccounts("accounts.txt");
        rff.printAccountList();
    }
}
```

- i) It appears that Exceptions are not handled. Modify the code to handle exceptions using the try-catch block and/or throws mechanisms. [4 marks]
- ii) Modify the code to add the account details to a HashMap where the keys represent account numbers and the values represent the corresponding balances (key:accno, value:balance). [4 marks]
- c) Using the following example, explain the concept of bounded generic types. [3 marks]

```

public class Box<T> {
    private T t;
    public <U extends Number> void inspect(U u) { % ... }
    public static void main(String[] args) {
        Box<Integer> integerBox = new Box<Integer>();
        integerBox.inspect("some text");
    }
}

```

d) Consider the program below.

```

class MalException1 extends RuntimeException {}
class MalException2 extends RuntimeException {}

public class Main {
    public static void main(String[] args) {
        String input = "Code A";
        try {
            System.out.println("Start Job");
            doJob(input);
        } catch (MalException1 se1) {
            System.out.println("Mal Function 1");
        } catch (MalException2 se2) {
            System.out.println("Mal Function 2");
        } finally {
            System.out.println("End Job");
        }
    }

    static void doJob(String input) {
        System.out.println("There is a risk");
        if ("Code A".equals(input)) {
            throw new MalException1();
        }
        if ("Code B".equals(input)) {
            throw new MalException2();
        }
        System.out.println("Perform Job");
    }
}

```

- i) What is the purpose of the finally block? [3 marks]
- ii) Determine the output of the program. [4 marks]
- iii) Can the program catch both exceptions (MalException1 and MalException2)? Explain your answer. [3 marks]

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